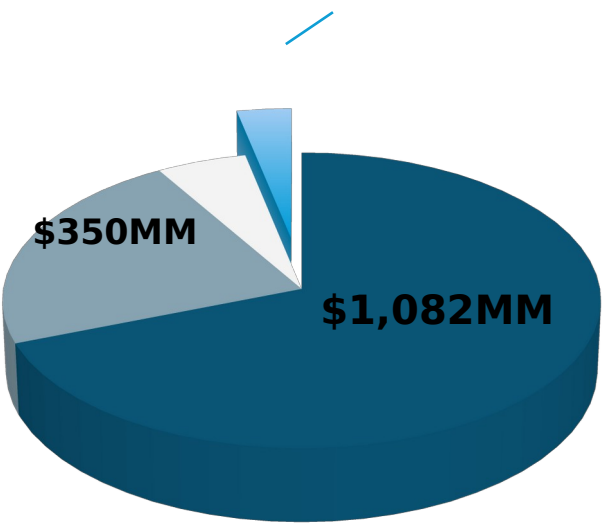




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Ethanol	\$518MM/yr
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Revenue	\$1,565MM/yr
Gross Profit	\$965MM/yr
Net Profit	\$714MM/yr
SPBP	3yrs
DCFROR	85%
ROI	114%

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Capital Cost	
ISBL	\$95MM
OSBL (40% ISBL)	\$38MM
ECC (10% ISBL)	\$10MM
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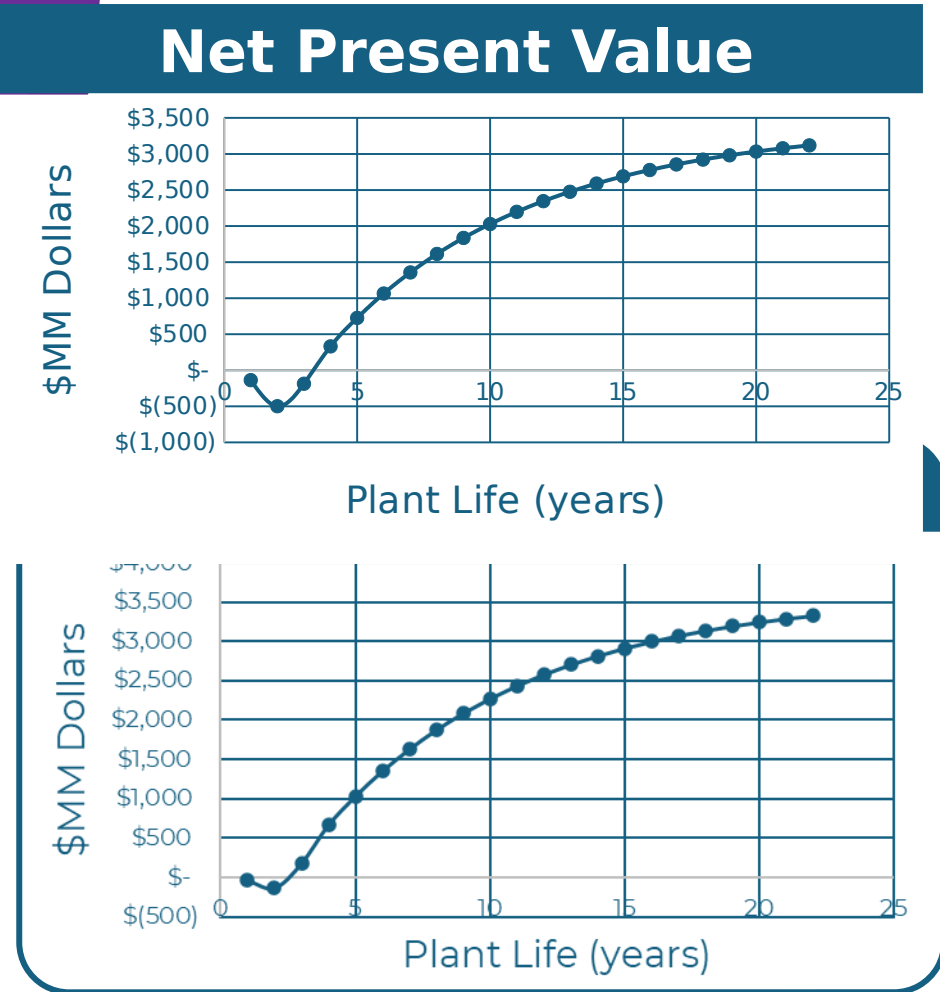
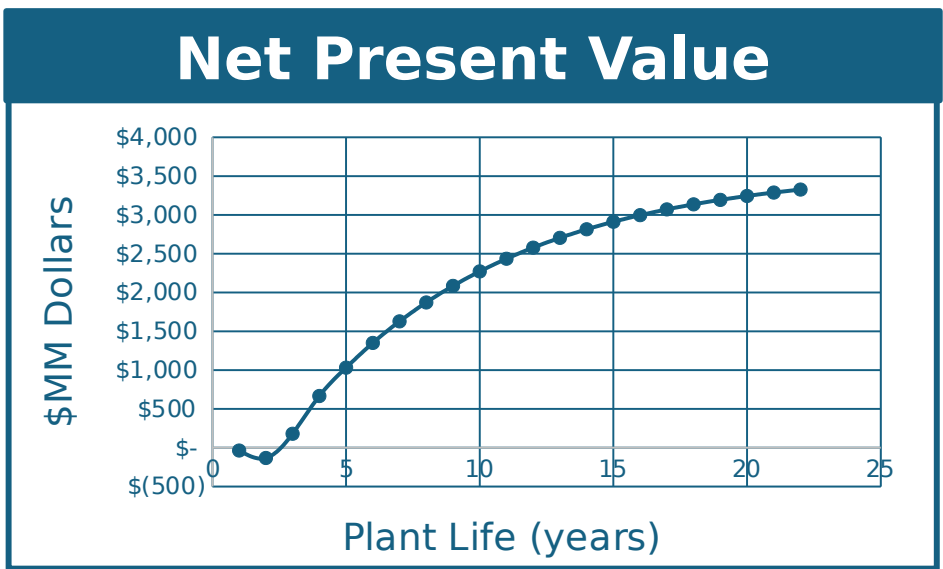
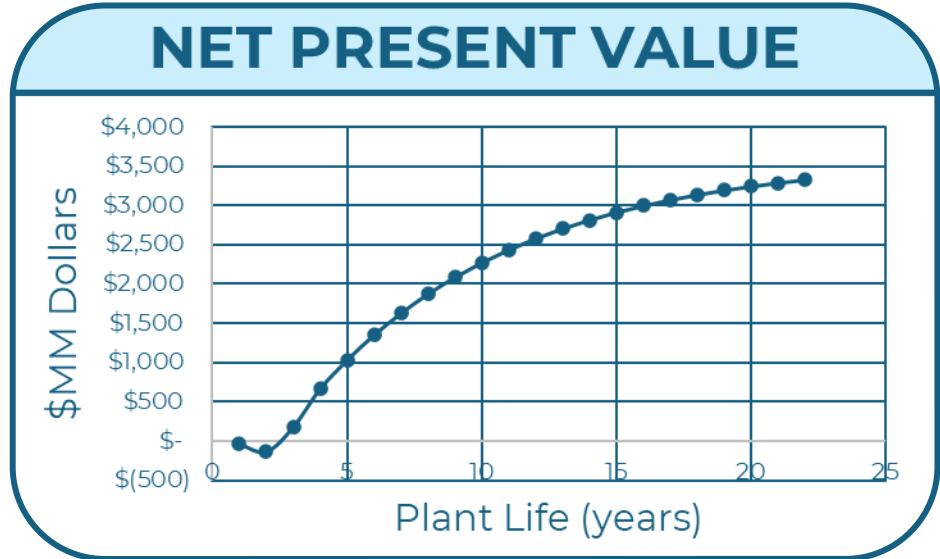
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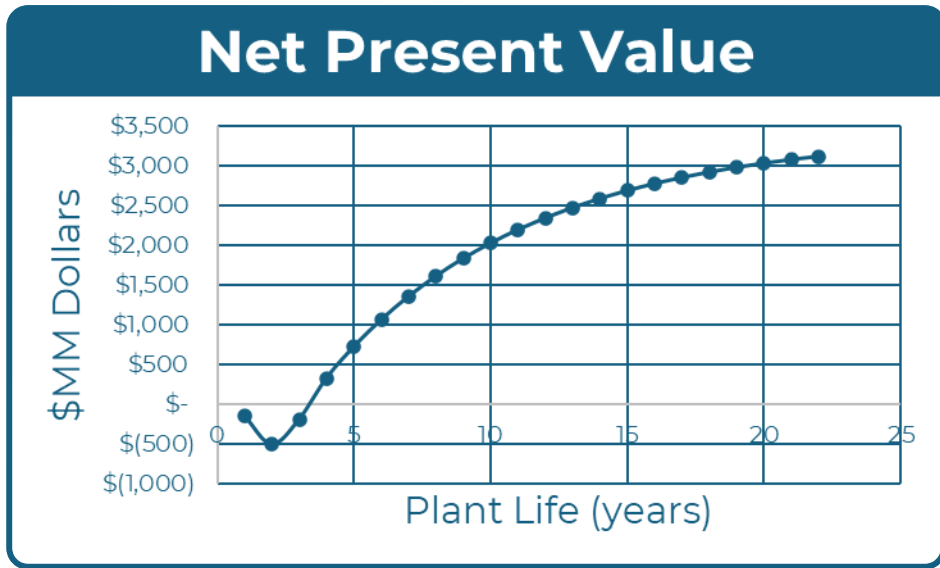
Financial KPI's	
SPBP	2.27yrs
DCFROR	219%
ROI	433%
NPV (at year 20)	\$3,555MM

Financial KPI's	
Revenue	\$1,565MM/yr
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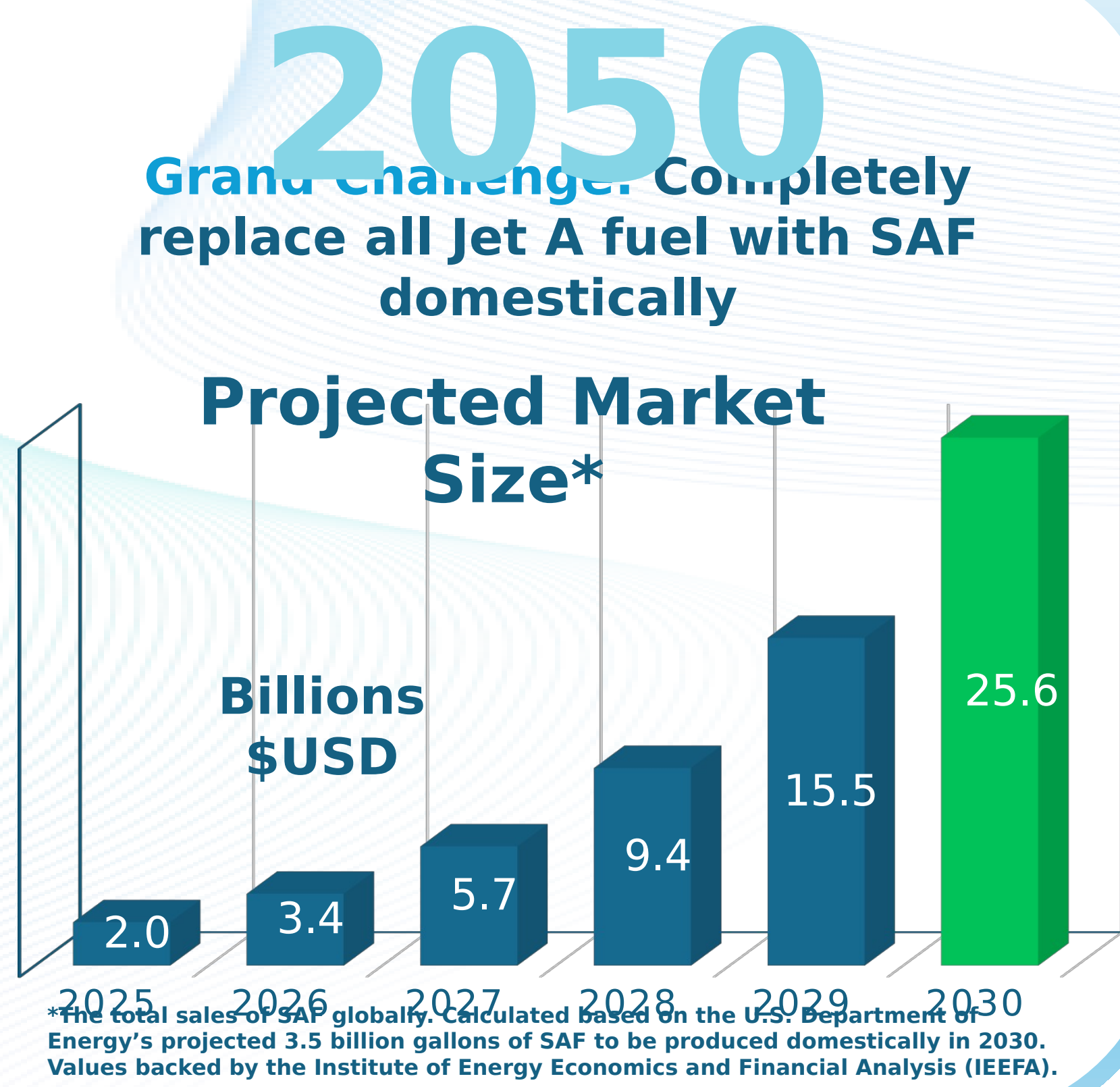
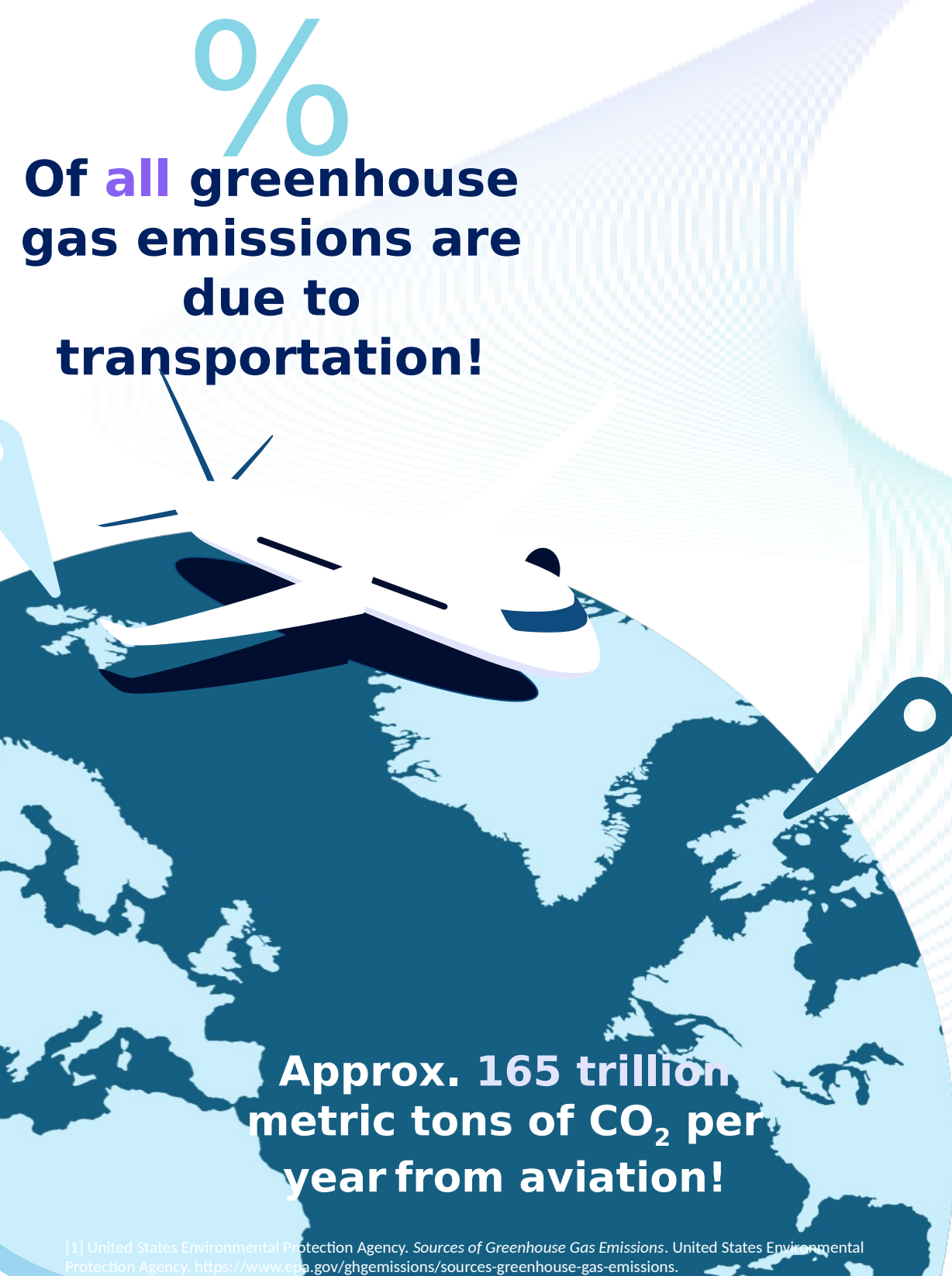


Ethanol- 122.53 tons/hr
Ethylene- 73.7 tons/hr
Water- 59.4 tons/hr
Olefins- 82.75 tons/hr
Jet fuel- 58.4 tons/hr
Hydrogen- 1.55 tons/hr

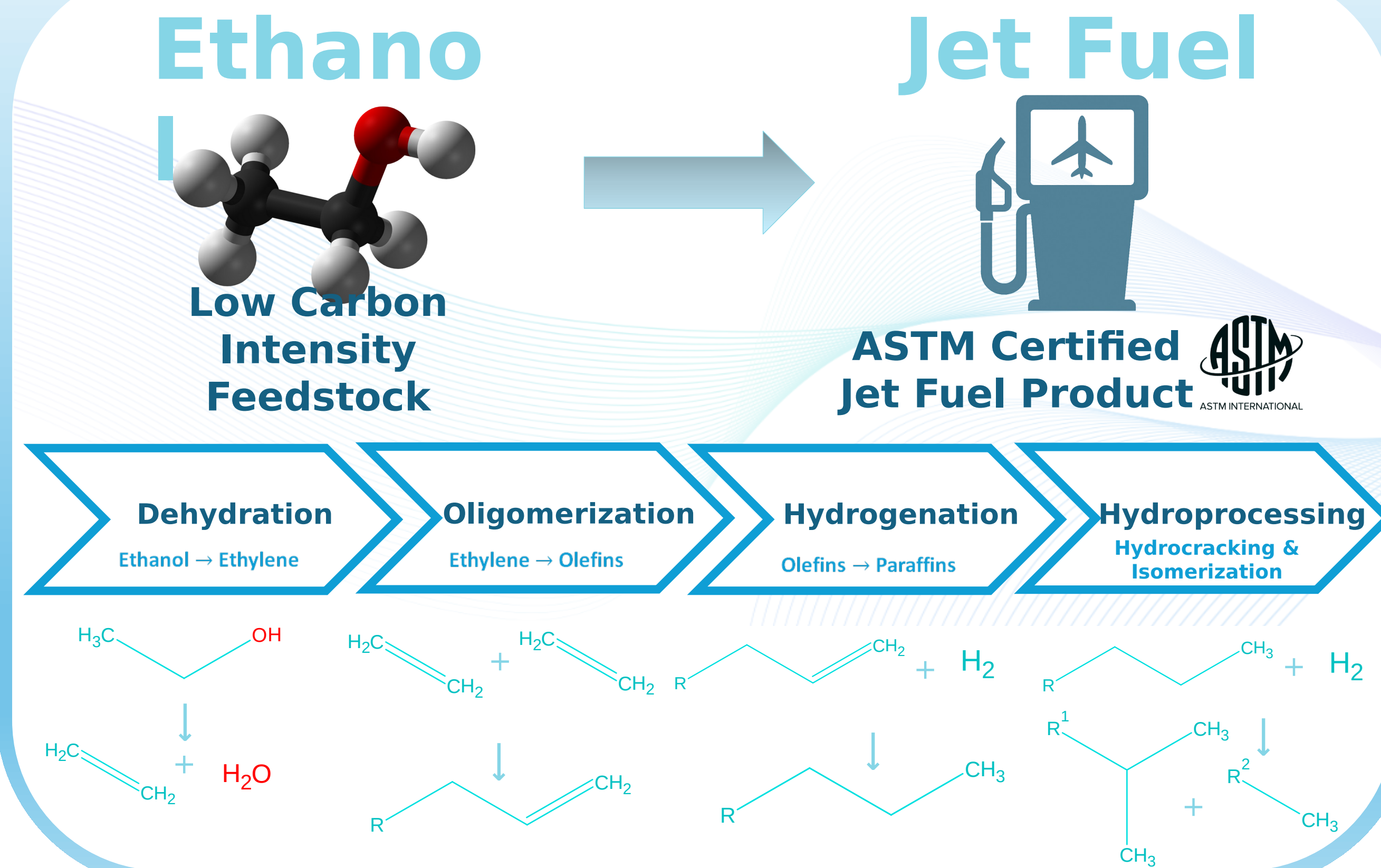
Clean Fuel, Cleaner Skies

Ethanol to Jet Fuel

The Climate Change Opportunity



Process Overview



Process Plant Design Basis

Location: Wilkinson County, Georgia
In-state feedstock & access to Port Savannah for international shipping!

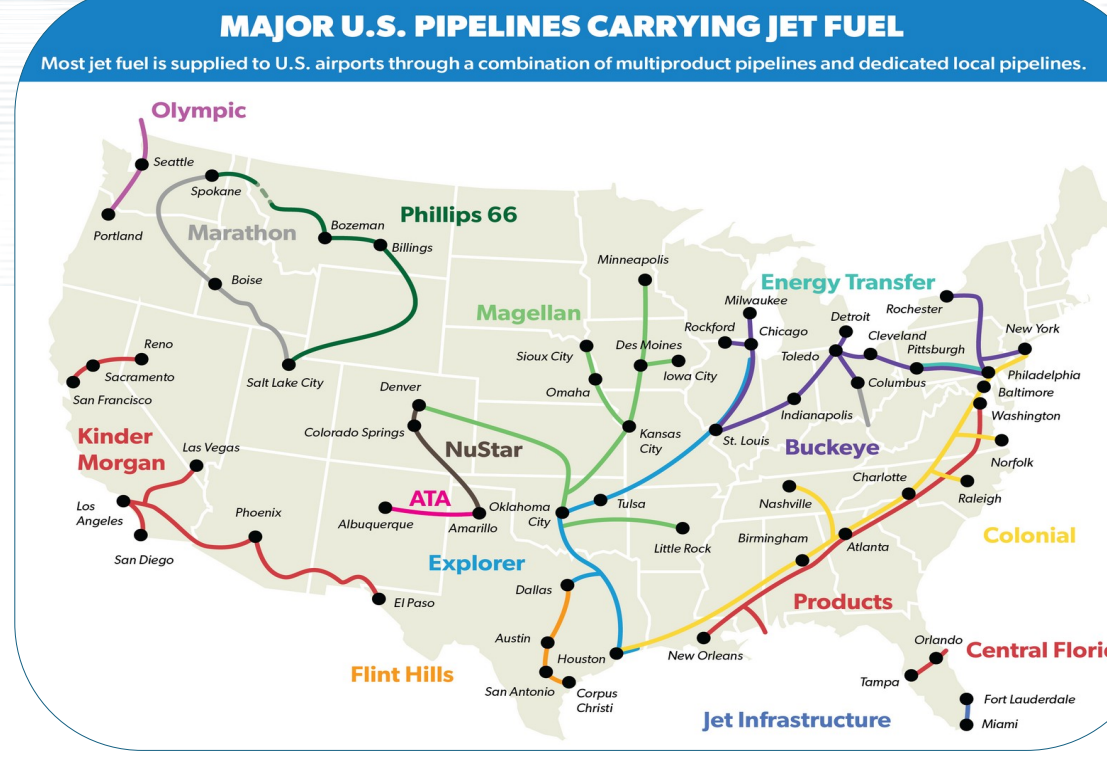
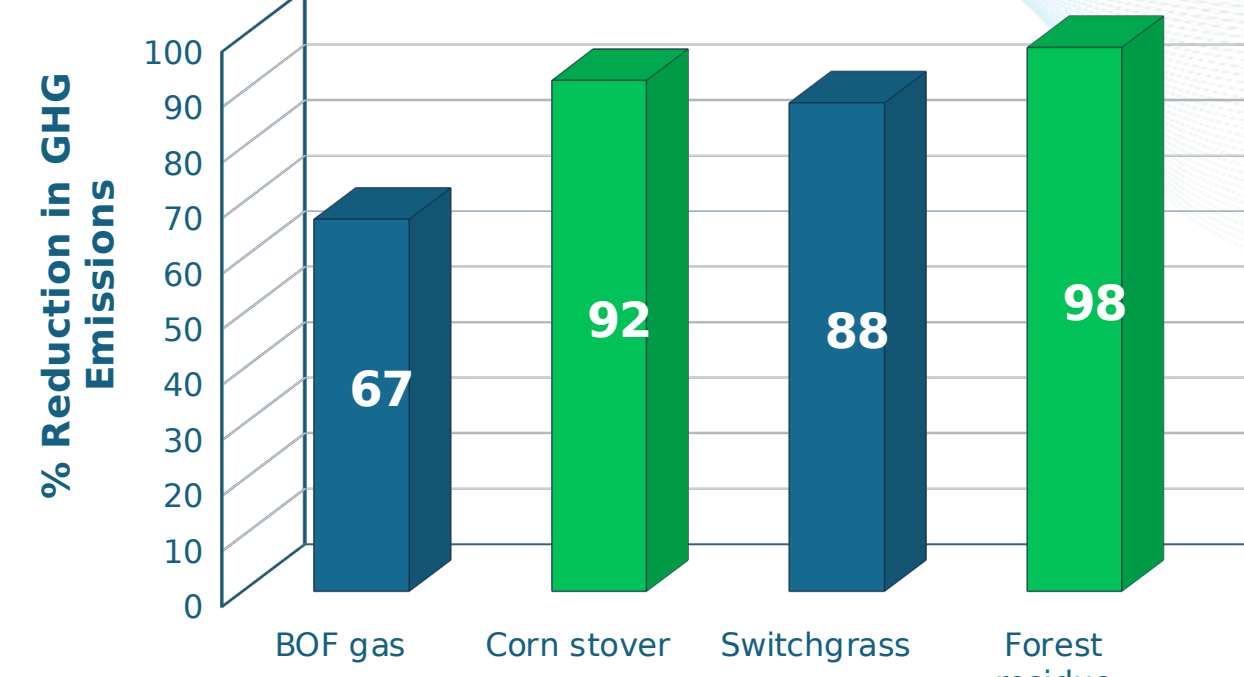
Lanza-Tech ethanol feedstock qualifies for **RINs & Tax Credits!**

Production: 10,000 Barrels/Day
This high daily production maximizes profitability.

Close proximity to the **largest airport**, Hartsfield-Jackson to sell SAF!



GHG Reductions from LanzaTech Bio-Derived Ethanol Feedstock

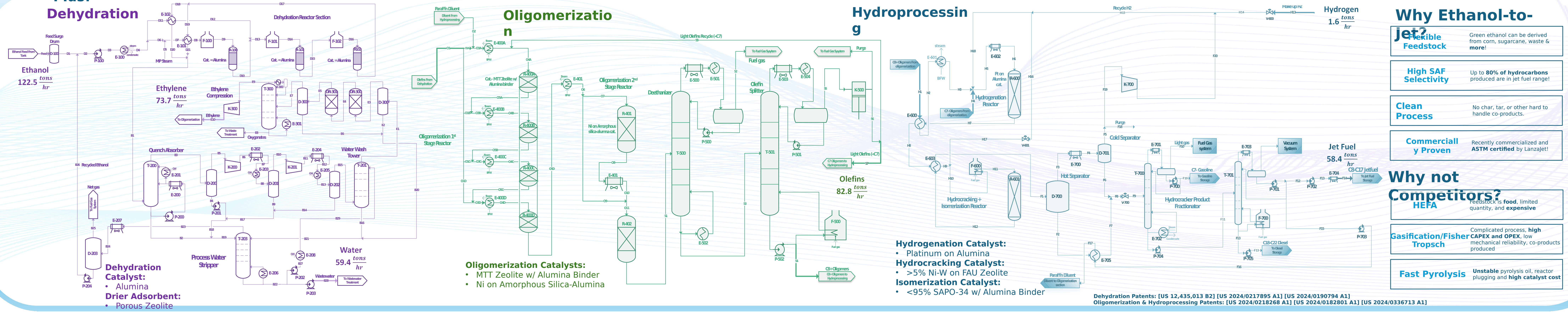


Fully Simulated & Converged w/ Aspen Plus!



Ethanol-to-Jet Process Design & Technology Comparison

Integrated Concepts from **Honeywell UOP** & **LANZATECH**



Process Safety

Safety

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Ethanol



Hydrogen



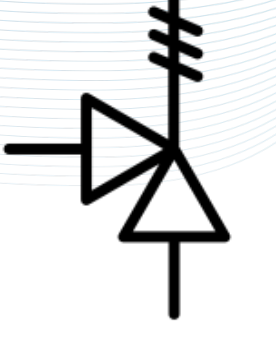
Ethylene



Petroleum Distillate



Most chemicals in the process are **flammable**.



PRVs safeguard against pressure and temperature excursions



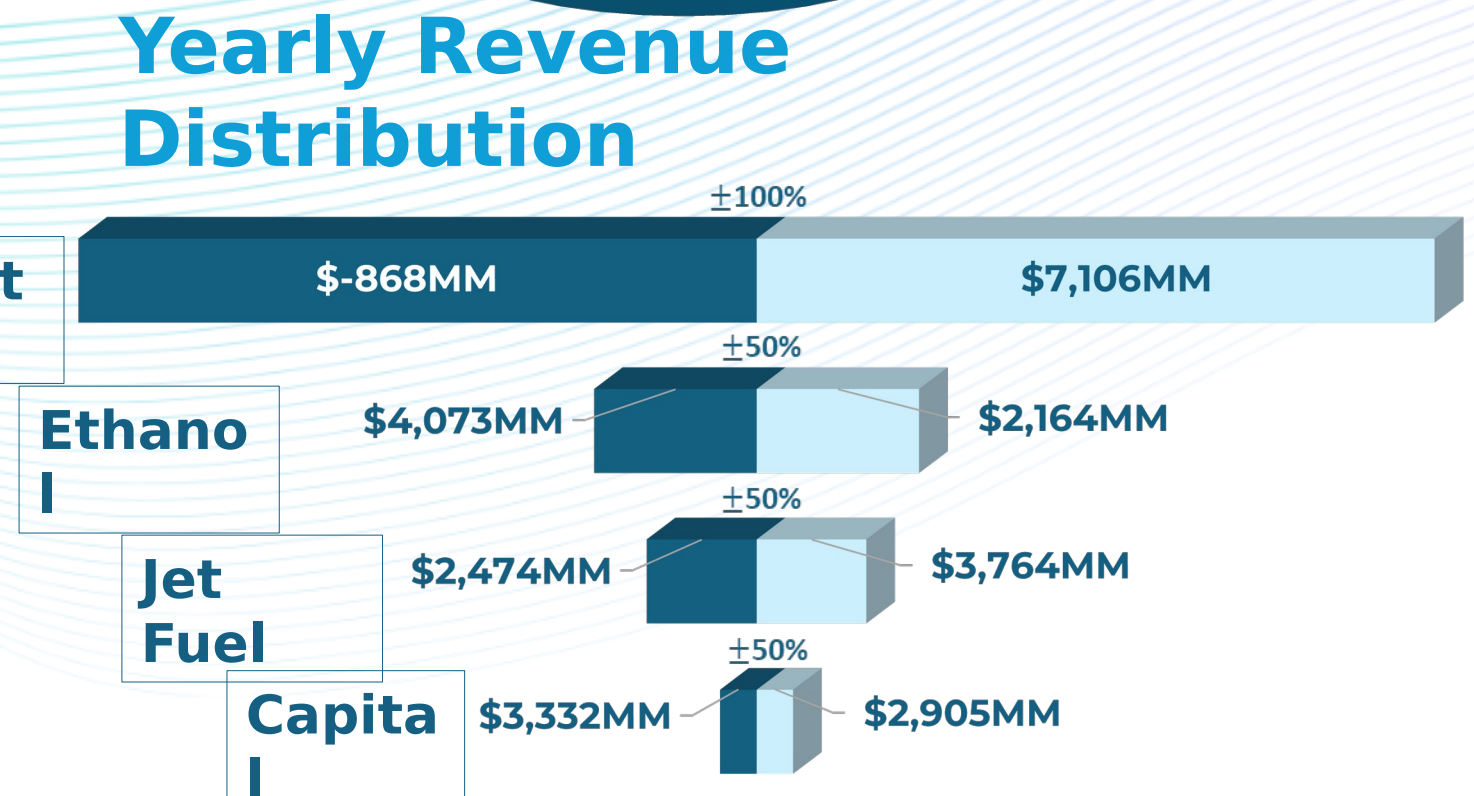
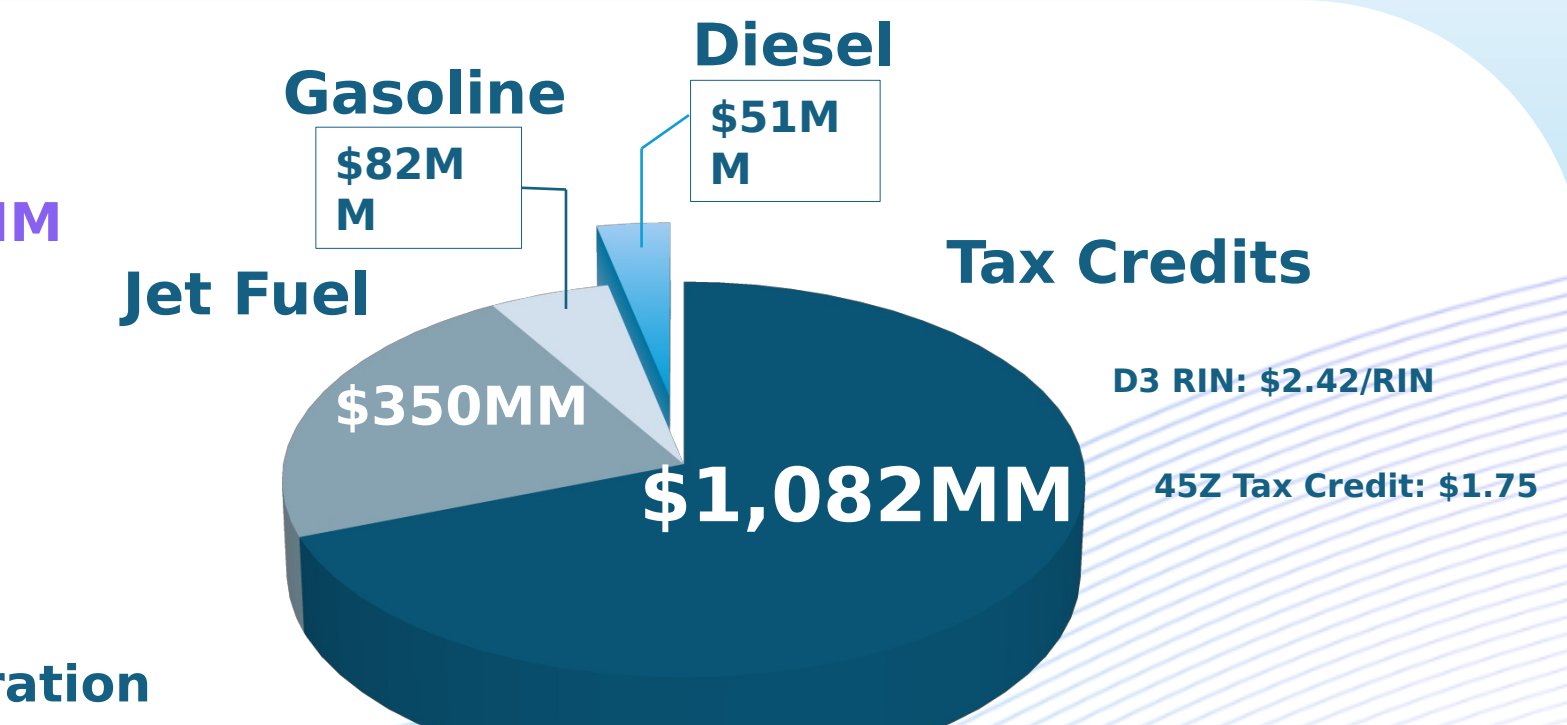
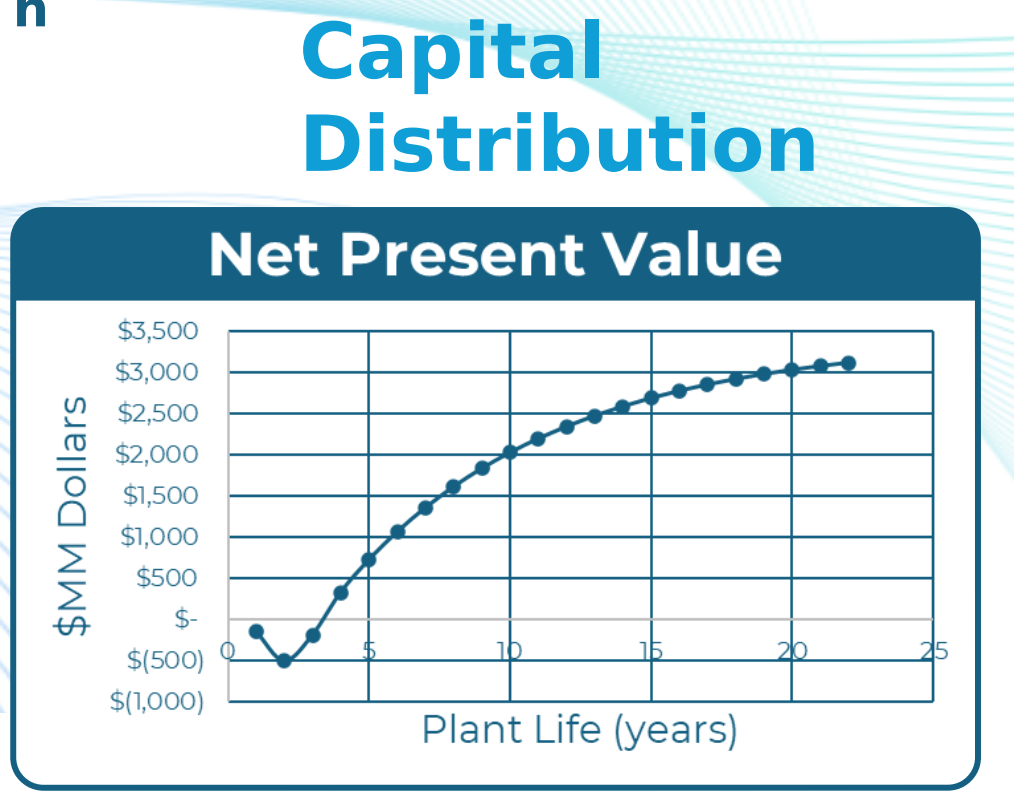
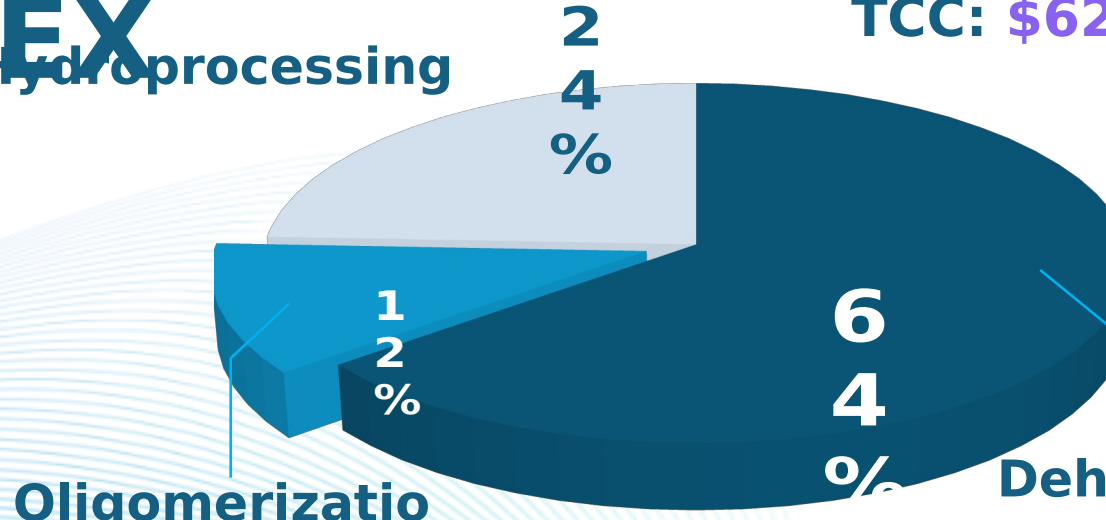
Personal protective equipment is required at all times.

Economic Analysis

OPEX & CAPEX

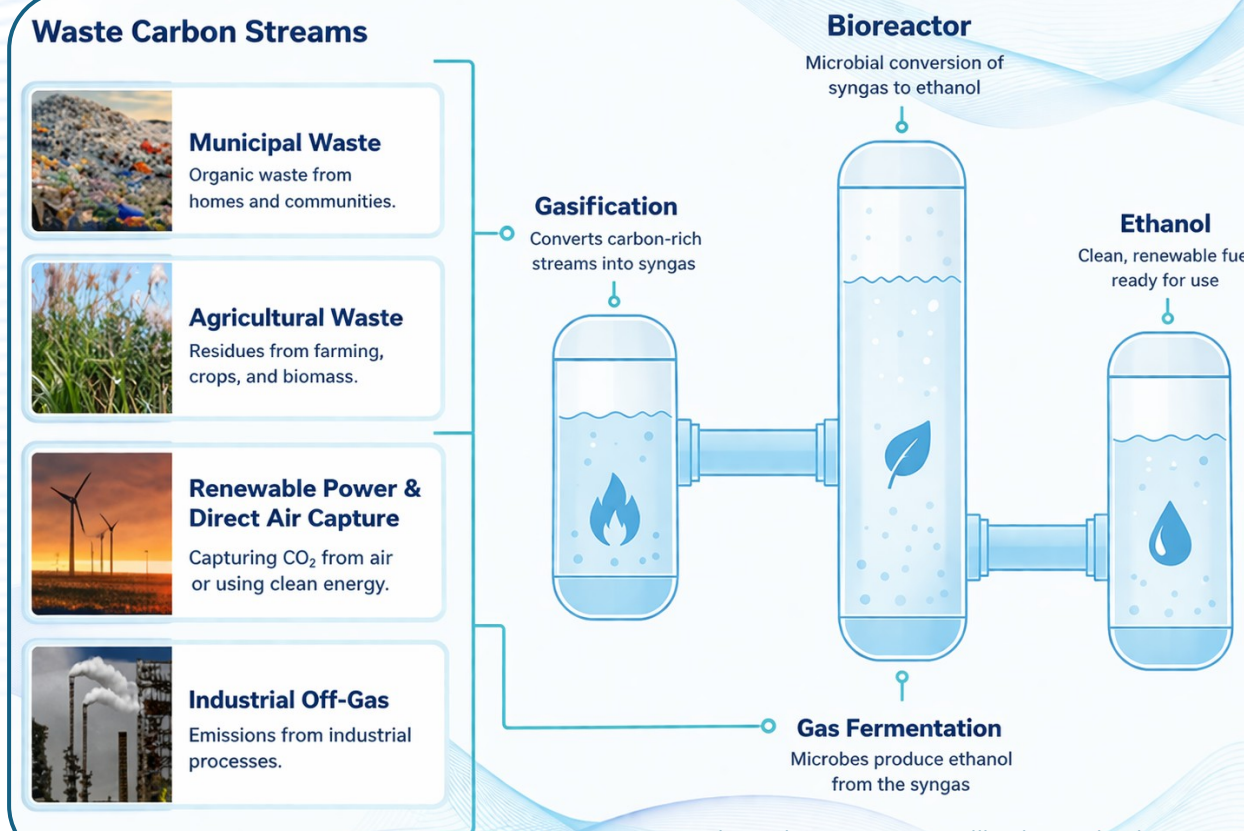
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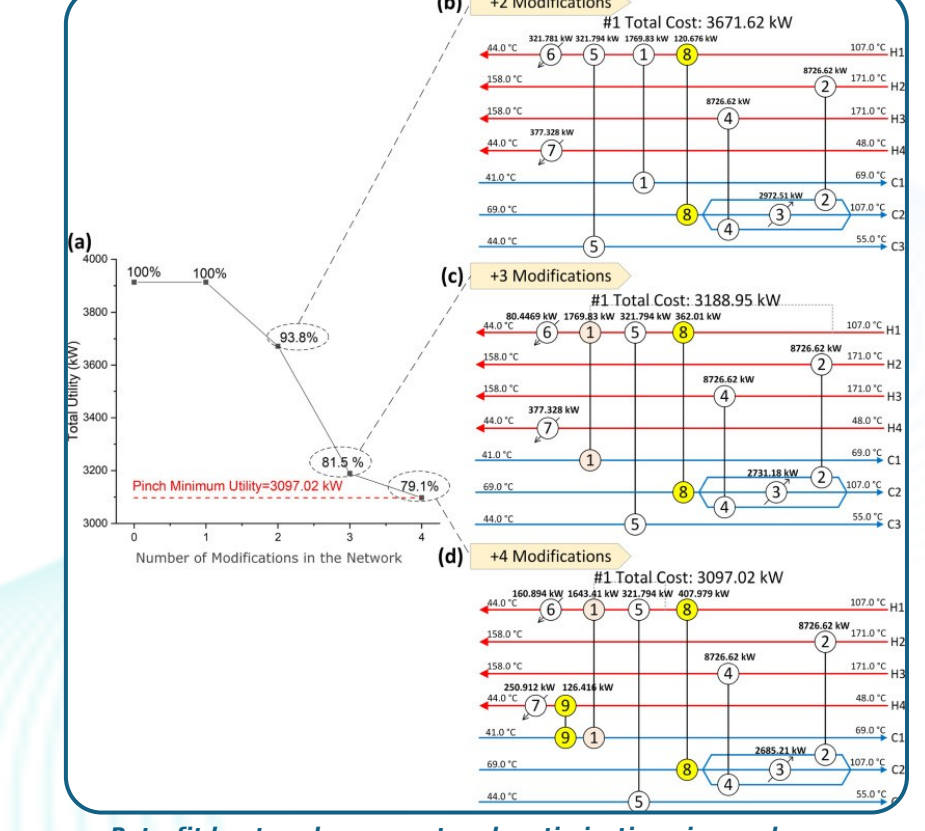


Recommendations

Integrate Low CI Ethanol & Hydrogen Production



Heat Integration from Oligomerization to Dehydration



Increase Design Capacity

day → day